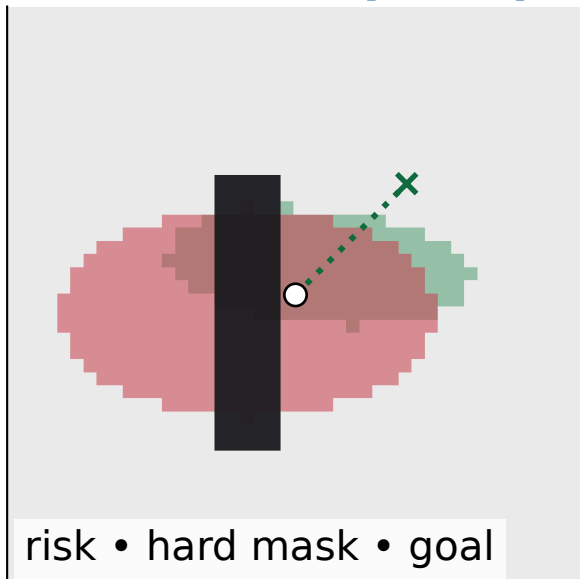


Material-aware port-Hamiltonian navigation

1 Local BEV percept



2 Feasibility witness

sample K short maneuvers

traversable + clearance

risk improvement $\Rightarrow g$

3 Additive energies

$$H = H_{\text{kin}} + H_{\text{geom}}$$

$$+ \lambda_s \tilde{r} + \lambda_h b(\phi)$$

heads predict λ_s, λ_h

4 Force channels

$$f_{\text{mat}} = -\nabla \tilde{r}$$

$$f_{\text{haz}} = -b'(\phi) \nabla \phi$$

$$F = f_{\text{geom}} + g \lambda_s f_{\text{mat}} + \lambda_h f_{\text{haz}}$$

5 Closed-loop update

semi-implicit p_{t+1}, q_{t+1}

gate affects material exposure

$g = 0$ recovers geometry-only

6 Tail-sensitive learning

roll out trajectory cost J

optimize $\text{CVaR}_\alpha(J)$

focus updates on worst rollouts